

Automated Sump Pump Control System

Problem

I developed this automated sump-pump system as a hands-on application of process control and industrial automation. The project provided a practical way to explore level sensing, relay logic, alarm handling, programmed sequences, and fail-safe design. The system was designed to maintain safe liquid levels automatically, reduce operator intervention, and respond appropriately to abnormal conditions, minimizing the risk of overflow and equipment damage.

Process

I selected the Arduino Opta, level sensors, pump, relays, and supporting hardware based on their function and system compatibility. I then modeled the modular frame and component layout in SolidWorks before fabricating and wiring the assembly. After programming the operating sequence in ladder logic, I tested multiple operating cycles to verify consistent pump operation, sensor response, and alarm behavior under high-level conditions, confirming that the mechanical, electrical, and control elements worked together reliably.

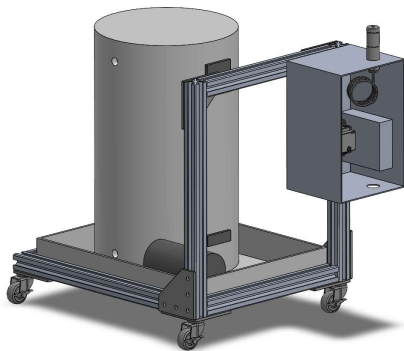


Figure 1: Full SolidWorks Assembly

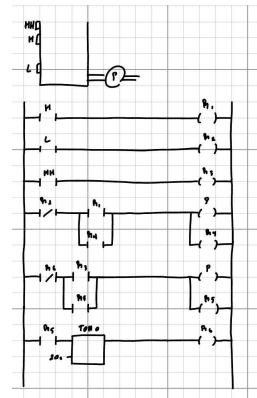


Figure 2: System Ladder Logic

Product

The completed system automatically starts the pump at the high-level sensor and shuts it down at the low-level sensor, maintaining the target operating range while preventing rapid cycling. A high-high-level alarm signals abnormal or failed-pumping conditions, while secondary containment limits overflow risk. A modular aluminum-extrusion frame supports the pump, sensors, controller, and process equipment. The finished system demonstrates discrete level sensing, relay-controlled outputs, programmed operating sequences, alarm handling, and layered fail-safe design in an integrated industrial automation application.



Figure 3: Physical Alpha Prototype

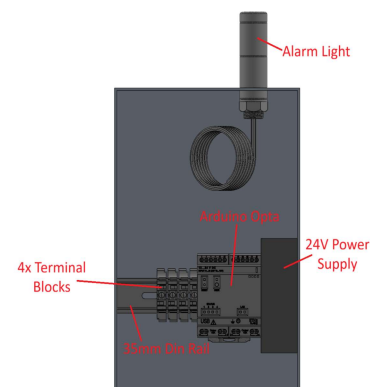


Figure 4: Electrical Box SolidWorks Assembly